Cloud Computing

* This uses a client-server architecture to deliver computing resources such as services, storage, databases & software over the Cloud.

* A cloud is stored on remote servers.
* It's managed by a cloud provider.
* We can access the cloud via anywhere in the internet.

Applications

→ Art
→ Business
→ Data Storage & Backup
→ Education.
→ Entertainment.
→ Management.
→ Social.

Advantages.

* Once the data is stored in the cloud, it is easier to get back-up & restore that ~~cloud~~ data using the cloud.
* Cloud applications improve collaboration by allowing groups of people to quickly & easily share information.
* Excellent Accessibility
* Low maintenance cost.

* Mobility => Cloud computing allows us to easily access all cloud data via mobile.
* Unlimited storage capacity.
* Data security.

Disadvantages

* Internet Connectivity.
* Vendor lock-in => This is the biggest disadvantage of cloud computing.
* Organizations may face problems when transferring their services from one vendor to another.
* Limited control.
* Security.
  -> While sending the data on the cloud, there may be a chance that your organization's information is hacked by hacker's.

-> Cloud computing uses a client-server architecture to deliver computing resources such as servers, storage, databases & software over the cloud.

-> Grid computing is also called as "distributed computing". It links multiple computing resources.

Characteristics : usage is monitored, measured.

→ Measured Service ⇒ ~~more~~
→ On-demand self service ⇒ can launch any cloud resource
→ Broad Network Access ⇒ can be accessed by any device
→ Rapid Elasticity ⇒ can be scaled.
→ Resource pooling ⇒ resources are shared among multiple users

Deployment Model

* This defines how & where cloud infrastructure is deployed & who can access it.

Types

1) Public Cloud

This is open to all to store & access info.

Ex:- AWS (EC2), Google App Engine.

* Low cost.
* Pay-as-you-go.
* Highly scalable.

2) Private Cloud

* This is known as internal cloud or corporate cloud.
* Used by organizations.

Ex:- Banks, governments.

Ex:- High security, Higher cost, Full Control.

3) Community Cloud

* This allows systems & services to be accessible by a group of several organizations to share the info b/wn organization & a specific community.

Ex:- Universities, government departments.

* Shared cost
* Limited users.

4) Hybrid Cloud

* Combination public & private cloud.

Ex:- Enterprises with variable work-loads, E-commerce websites.

* Security
* Flexibility.
* Cost-Effectiveness.

Cloud service Models

1) Saas (Software as a Service)

→ This is a Software distribution model hosted by a vendor or service Provider.

* On-Demand Software.
* We just use the application.

Ex:- Microsoft 365, Salesforce.

* Low cost
* Less hardware
* Less maintenance cost.

2) Paas (Platform as a service)

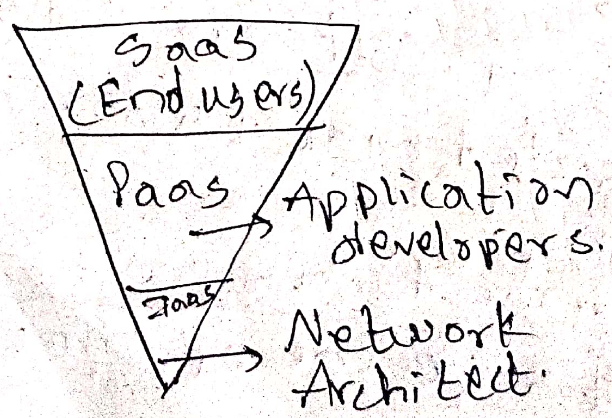
* The provider gives a ready-to-use platform to build & deploy applications.
* Focuses on coding, users have to code to develop.
* Generally rents the platform.

Ex:- Google Apps Engine (GAE), Windows Azure

* Faster App development.

3) Infrastructure as a service (Iaas)

* Offers the basic infrastructure like
  → Virtual machines
  → Storage
  → Networking.
* Rents the hardware



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RAAS ⇒ Recovery as a Service (or)
Disaster Recovery as a Service (DRaaS).

* Here a third-party provider manages the replication & hosting of an organization's physical or virtual servers to provide failover in cases of cyberattacks, or equipment failure.
* Goal is to enable rapid restoration.

BaaS (Backup as a Service)

* This focuses only on backing up & restoring data files.
* Requires manual effort to restore the IT infrastructure.
* This restores only data, whereas applications, O.S. & servers must be fixed or rebuilt by the client.

Note:- DRaaS restores the entire system not just the data.

DRaaS Provider Models ⇒ Self-service DRaaS, Assisted DRaaS, Managed DRaaS.

IDAAS (Identity as a Service)

* Allows users to connect to & use identity management services from the cloud.

Core functions
* A data stores
* Query Engine.
* Policy Engine.
* Identity management ensures the right people in an organization have the right access to the right resources.
→ Delivers access services efficiently & cost-effectively.
→ Protect against internal & external security threats.
→ Meet regulatory compilance requirements around security & privacy.

Components of IDaaS

1) Cloud based & multi-tenant architecture
* Provides lots of benefit such as the vendor can issue updates, security fixtures & improves performance.

2) Security.

3) Single Sign-On Federation
* This allows a user to log in once & gain access to multiple applications within the same organization without logging in again.

* Federation enables users to access external applications using credentials from a trusted identity provider.

4) Analytics & Intelligence

5) Governance, risk & Compilance.

STaas (Storage As a Service)

* This is a third party provider offers organizations access to data storage capacity & related management services on-demand.

Types

1) Object storage
2) File storage
3) Block storage.

* Can be accessible from anywhere

* Scalability.

Maas (Monitoring as a Service)

* Third party provider remotely monitors the health, performance & security of an organization's IT infrastructure, applications & processes.

→ AWS = Largest, wide services.
→ Azure = enterprise focused
→ GCP = Strong in AI & analytics.
→ OCI = infrastructure based.
→ IBM cloud.

AWS

This offers an extensive range of services that cater to almost every possible use case, from infrastructure management to machine learning & beyond.

Compute

* EC2 (Elastic Compute Cloud) = Virtual servers.
* Lambda = Serverless computing without provisioning servers.
* ECS (Elastic Container Service)

Storage

* S3 (Simple Storage Service)
* EBS (Elastic Block Store)
Block-level storage for EC2 instances

Databases

* RDS (Relational Database Service) :-
(MySQL, PostgreSQL, SQL Server).
* DynamoDB :- Managed NoSQL database

Networking

* VPC (Virtual Private Cloud)
* Cloudfront => Content Delivery Network to distribute content globally with low latency.

Machine Learning & AI

* SageMaker
* Lex

Analytics

* RedShift
* Athena

Microsoft Azure

* This is a popular cloud platform for especially enterprises using Microsoft technologies.

Compute

* Azure Virtual Machines
* Azure Functions.

Storage

* Azure Blob Storage
* Azure Disk storage.

Databases

* Azure SQL Database
* Cosmos DB.

Networking
* Virtual Network
* Azure CDN

Machine Learning & AI.
* Azure Machine Learning.
* Cognitive Services.

Analytics
* Azure Synapse Analytics
* Power BI.

GCP (Google Cloud Platform)
* This is known for it's high-performan compute services & innovative tech-nologies in data & machine learning.

Compute
* Google Compute Engine
* Google Kubernetes Engine.

Storage
* Google Cloud Storage.
* Persistent Disks

Databases
* Cloud SQL
* Bigtable

Networking
* Virtual Private Cloud
* Cloud CDN.

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Machine Learning & AI

* AI platform
* Auto ML

Analytics

* BigQuery.
* Dataflow.

IBM Cloud

* This is particularly strong in hybrid cloud environments, AI, & blockchain.

Compute

* Virtual Servers
* kubernetes.

Storage

* Cloud Object Storage.
* Block Storage.

Databases

* IBM Db2
* Cloudant.

Networking

* IBM Cloud CDN
* Virtual Private Cloud (VPC).

Machine Learning & AI

* Watson AI
* Watson Studio.

Blockchain

* IBM Block chain.

→ AWS has maximum number of region & availability zones.

* Facilities on almost on every continent.

→ Azure has very strong presence especially in Microsoft dominated areas.

→ GCP is the AI/ML & analytics leader

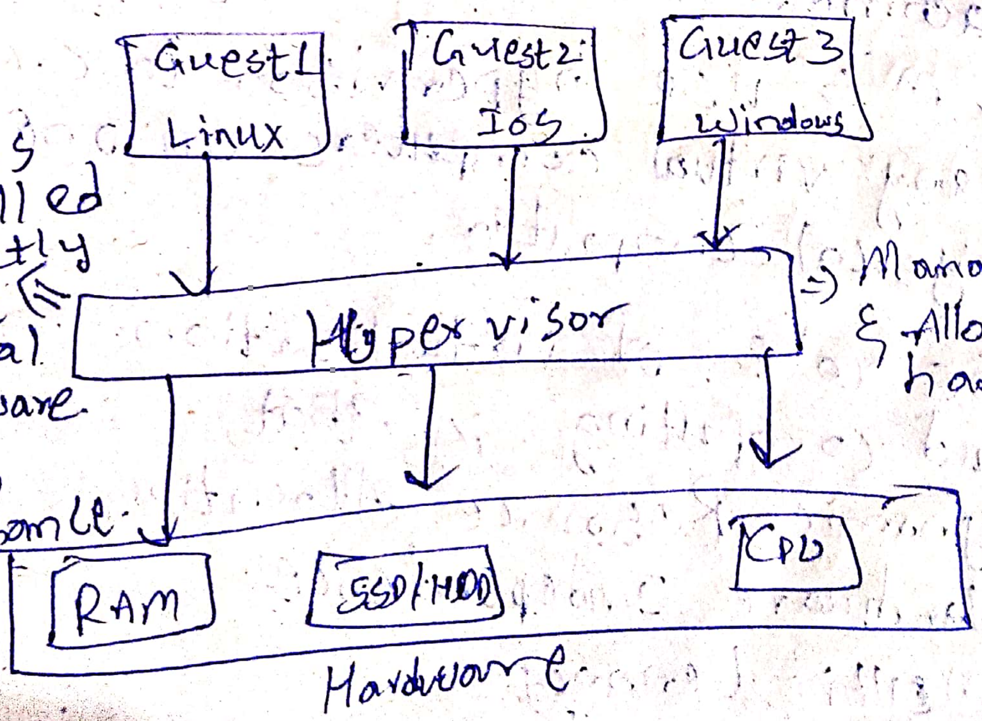
CO-2

Virtualization

* This is a process that abstracts physical hardware resources & presents them as logical ; isolated computing environments.
* Virtual machine => A self contained system with it's own virtual CPU, memory, storage & network interface.
* This allows to create multiple simulated environments.
* By this the CPU can be utilized efficiently.

Types

1.) Bare - Metal

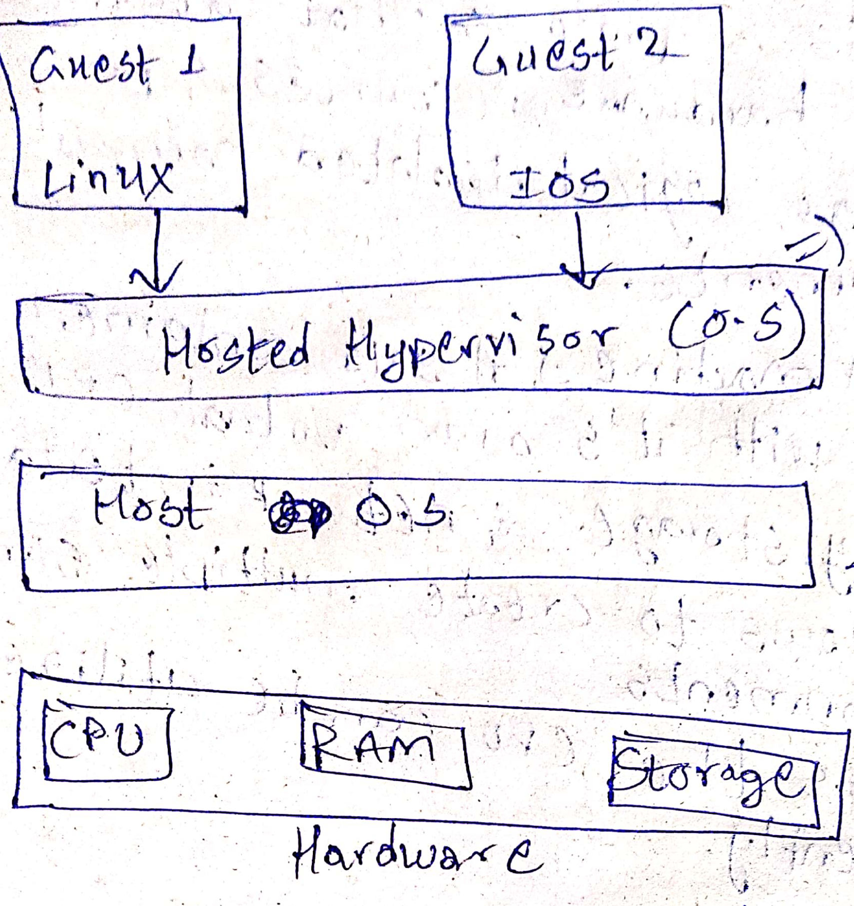


This is installed directly on the physical hardware.

* High Performance.

=> Manages & Allocates hardware

2.) Hosted



=) Installed as an application on top of an existing O.S.

*Low Performance

* Virtual machines behave like actual computers but use the same physical machine.
* VM's use hypervisor to create many virtual computers on one physical computer.
* The role of virtualization in cloud computing is that,
1) Dynamic Resource Allocation.
2) Hardware Independence.
3) Multi-tenancy.
* VM's are also called as cloud instances.

* Real physical Computer = Host.
* Vm's = Guests.
* A virtual machine O.S can be different from host O.S (or) can be the same.

Types of Virtualization

→ Application
→ Network
→ Desktop
→ Storage
→ Server
→ Data.

Application Virtualization

* Applications run in an isolated environment without being fully installed on the local system.

Ex:- Vmware.

Network Virtualization

* Abstracts physical network resources into logical, software-defined networks.

Ex:- VLAN.

Storage Virtualization

* Combines multiple physical storage devices into a single logical storage pool.

Ex:- SAN.

Server Virtualization

* A physical server is divided into multiple vm's by hypervisor.

Ex:- KVM.

Data Virtualization

* Provides a unified view of data from multiples sources with-out physically moving it.

Ex:- IBM Data virtualization.

Virtualization	Cloud Computing
* Creates virtual versions of hardware	* Provides services or computing resources via the internet.
* This is a tool.	* This is a service.
* We own/manage the hardware	* We rent resources, the provider owns the hardware.

Characterstics

→ Partitioning.
→ Isolation.
→ Encapsulation.
→ Hardware Independence.
→ Resource Pooling.
→ Flexibility & Scalability.
→ Improved Utilization
→ Cost efficiency.
→ Fault Tolerance & High Availability.

* Virtualization are also classified on the type of resource being virtualized

Hardware Virtualization

* Creates Virtual machines that emulate (copies) physical hardware.

- Full
- Para
- Hardware Assisted

} virtualizations.

Operating System Virtualization

* Allows multiple isolated user space instances to run on a single O.S kernel.

File-level Storage virtualization

* Provides a unified file system over multiple storage locations.

Desktop Virtualization

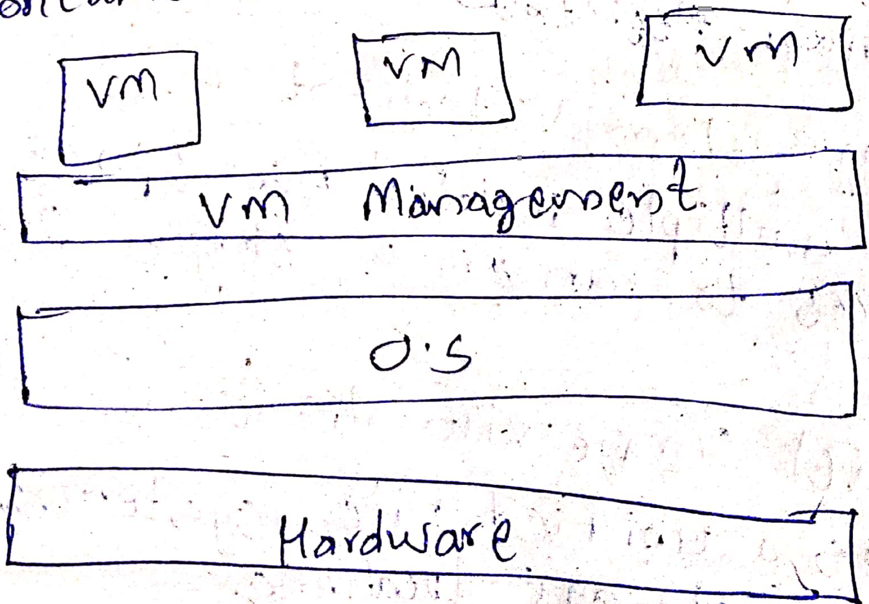
* Provides users with virtual desktops hosted on centralized servers.

Hardware Virtualization products

* Vmware ESXi
* Microsoft Hyper-V
* Xen

O.S Virtualization

* This is a server virtualization technology that includes customizing a conventional operating system to run programs for many users on the same ~~cha~~ machine simultaneously.
* The main operating system in O.S virtualization behaves like numerous other independent sources (or) systems.
* O.S virtualization is also called as ~~Con~~ Containerization.



Types of O.S Virtualization

* Windows O.S virtualization :) In this we use vmware Workstation software.
* Linux O.S Virtualization => Vmware workstation software is used.

→ O.S virtualization utilizes two virtual drives, so that client will be able to access that disc across network.

1.) Shared Disk.

* This can be used by several companies/clients simultaneously.

2.) Private Disk.

* This can be only used by one firm or client.

Features

→ Efficient Resource Utilization.
→ Rapid Deployment.
→ Process-Level Isolation
→ High Portability.
→ Microservices support.
→ Strong Ecosystem Support.

Working Process

* O.S virtualization operates at the kernel level, ~~to~~ also gains ~~the~~ (or) makes the host O.S's kernel to creates manage isolated enviroments.

Containerization

* Containers are lightweight, portable, & self-contained units that encapsulate & it's dependencies.
* Each container shares the host O.S's kernel but maintains it's own isolated filesystems & process space.
* O.S virtualization utilizes Linux namespaces to provide process-level isolation between containers.
* Container runtimes, such as Docker or containerd, are responsible for creating, managing, & executing containers in the host system.

→ Orchestration platforms like Kubernetes provide higher-level tools for managing containerized applications at scale.

Para Virtualization

* This is a type of hardware virtualization that enables the guest O.S in a VM to access the hypervisor directly.

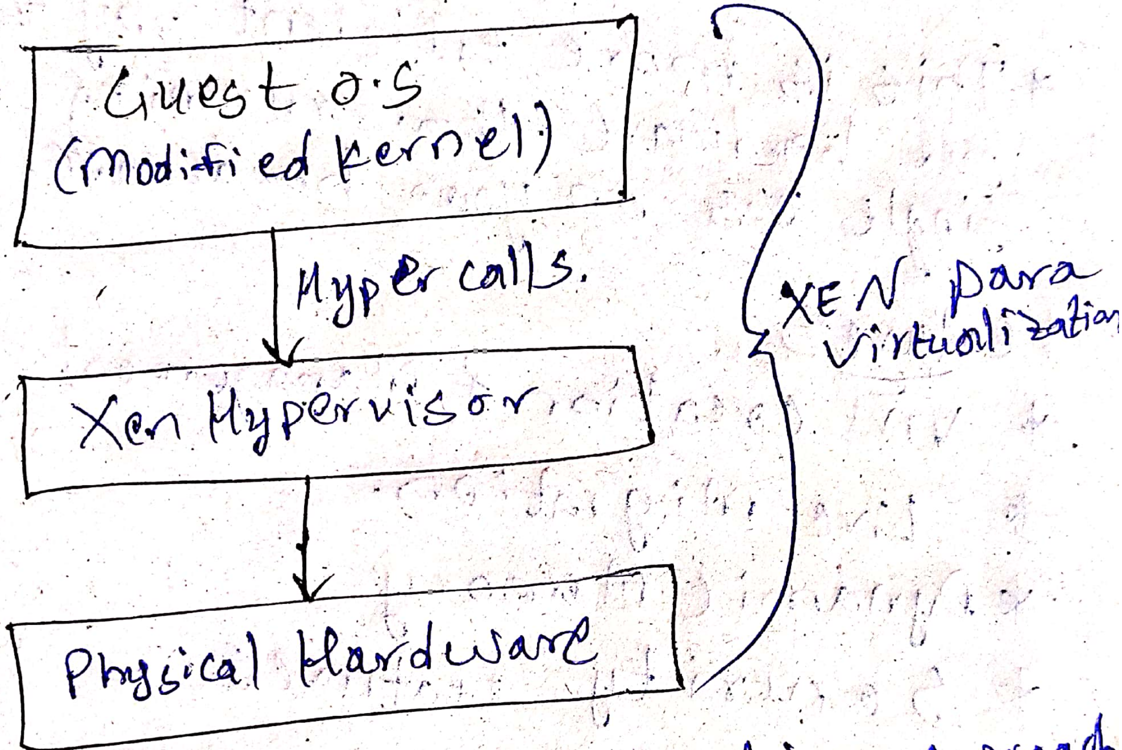
Ex:- Xen (hypervisor) for paravirtualization

Xen (Para-Virtualization)

* This enables high performance to execute guest O.S.

Physical Hardware → Xen Hypervisor → Host Domain → Guest Virtual Machines

* Xen run directly on the hardware, not on a host O.S.



* The above is a virtualization approach where the guest ~~oper~~ O.S is modified to interact directly with the Xen hypervisor.

VMWARE (Full-Virtualization)

* Here the guest O.S's are unmodified in running. This is the major benefit.

* Binary translation is portable for full virtualization.

* This delivers the best isolation & security for Virtual Machine.

Hyper - Virtualization

* Hyper -V is a hardware product from Microsoft.
* Each Vm works like a computer with an O.S & running programs.
* This is more efficient way to use our hardware than running a single O.S on your hardware.

Features:

* Vm creation & Management
* Live Migration.
* Dynamic Memory
* Security Features.

→ Supports: Linux, Windows, Free BSD,
macOS is not supported.

→ VMware supports macOS, Windows
Linux & unix operating systems.